

SYNEX GROUP · SYSTEM HANDBOOK · OCTOBER 2026

Synex QB Portal

A web portal that puts QuickBooks Desktop 2024 in the browser for every Synex Group company: read all lists and transactions, create and edit them with approval, run reports, and manage users and access from one place.

Address **qb.synexint.com** QuickBooks record types **50** QuickBooks reports **36** Automated tests **434**

Sync interval **1 minute**

What the system does

QuickBooks Desktop runs on one Windows computer and only people sitting at that computer can use it. The Synex QB Portal keeps a live copy of the QuickBooks company file on a web server, so staff can work from any browser or phone while QuickBooks stays the system of record.

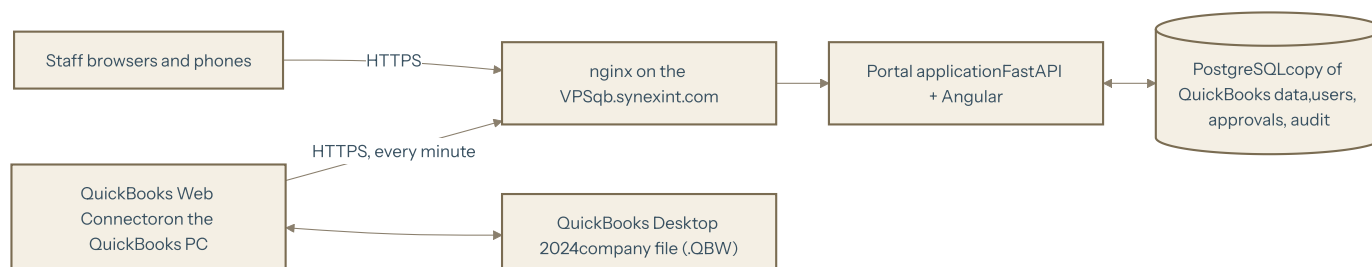
- **Read everything:** all customers, vendors, items, accounts and transactions, refreshed from QuickBooks every minute.
- **Write safely:** create, edit, delete and void records. Each change waits for approval (unless the user has direct-write rights) and is then written into QuickBooks on the next sync.
- **Work faster than in QuickBooks:** search across everything, aging, customer and vendor overviews, reorder alerts, bulk import from Excel, bulk PDF and e-mail, saved views.
- **Control access:** roles per module, per-company access, two-step sign-in and a full audit log.

QuickBooks remains the master.

The portal never changes QuickBooks data on its own. Every write is a QuickBooks request that QuickBooks itself validates; if QuickBooks rejects it, the change shows as failed with QuickBooks' own message.

HOW IT WORKS

QuickBooks Desktop cannot be reached over the internet directly. Intuit's **QuickBooks Web Connector** (QBWC), installed on the QuickBooks computer, calls the portal every minute over HTTPS and asks "is there any work?". The portal answers with qbXML requests (read changes, write approved changes, run reports) and QBWC passes the answers back.



One sync cycle

STEP 1

Connect

Web Connector signs in with the company's Web Connector user and password.

STEP 2

Check the file

The portal confirms the open company file is the one registered for that company, otherwise it stops.

STEP 3

Read changes

All record types modified since the last sync are read in one batched request.

STEP 4

Write

Approved changes are sent to QuickBooks one by one; each result is saved.

STEP 5

Reports

Reports that users asked for are run and stored.

The first sync of a company reads everything (transactions from the last 365 days). After that each sync reads only what changed, so a normal sync takes a few seconds. Deleted QuickBooks records are detected and hidden. Features that a company file does not have (for example multi-currency) are detected automatically and switched off for that company.

QuickBooks modules

All 50 record types can be created from the portal. 37 of them can also be edited; the other 13

cannot be edited through Web Connector (QuickBooks' own limit), so they are deleted and re-created instead.

MODULE	TYPES	RECORD TYPES
Sales & customers	9	Customers, Estimates, Sales orders, Invoices, Sales receipts, Credit memos, Received payments, Statement charges, Time tracking
Purchasing & vendors	7	Vendors, Purchase orders, Item receipts, Bills, Bill payments, Vendor credits, Bill payments by card
Banking	6	Checks & payments, Credit card charges, Deposits, Credit card credits, Transfers, Vehicle mileage
Accounting	7	Chart of accounts, Classes, Payment terms, Payment methods, Journal entries, Sales tax codes, Currencies
Inventory	2	Inventory adjustments, Assembly builds
Items & lists	19	Inventory, Non-inventory, Service, Other charge, Discount, Subtotal, Payment, Sales tax, Group and Assembly items; Customer, Vendor and Job types; Shipping methods, Other names, Employees, Sales reps, Price levels, Vehicles

Reports

36 QuickBooks reports run inside QuickBooks itself, so the numbers match QuickBooks exactly: 14 general reports (Profit & Loss, Balance Sheet, Trial Balance, customer and vendor balance summaries and more), 17 detail reports and 5 aging reports. Results can be exported to Excel.

Workflows that follow QuickBooks

- Estimate or sales order → **Create invoice** (pre-filled).
- Purchase order → **Receive items** → item receipt → **Convert to bill**.
- Received payments and bill payments pick only open invoices or bills of the chosen customer or vendor.
- Custom fields from QuickBooks are shown on each record (read only).

Portal features

AREA	WHAT PEOPLE CAN DO
Home	Receivables, payables, this month's sales against last month, reorder count and approvals waiting; a 6 or 12-month sales and purchases chart; top 5 customers and vendors; recently opened records; one-click New invoice / customer / bill.
Search (Ctrl K)	Find any customer, invoice, item or page from anywhere, by name or number; start a new record from the same box.
Lists	Search, Active / Inactive filter, date buttons (This month, Last month, Last 3 months, This year), sortable columns, total of the filtered amount, choose columns, saved views, tick rows for Excel, PDFs in a ZIP, or e-mail each document to its customer.
Customer and vendor page	Owes / owed, overdue, credit limit, total sales, open invoices or bills with an aging bar, 12-month trend, latest payments, Receive payment / Pay bills.
Aging	Who owes us and whom we owe, split into Not due, 1-30, 31-60, 61-90 and over 90 days, per customer or vendor.
Items to reorder	Inventory at or below its QuickBooks reorder point, suggested order quantity, and one click to a draft purchase order for the preferred vendor.
Forms	Choosing a customer fills terms and addresses and warns when the credit limit is exceeded; choosing an item fills description, price or cost and shows stock on hand. Unsaved forms are kept as drafts in the browser. A new transaction with the same number, or the same name and amount within a week, asks before saving.
Documents	Printable PDF of invoices, estimates, orders, receipts and more with the company logo; e-mail with the PDF attached; file attachments on any record (stored by the portal, not in QuickBooks).
Import	Bulk create from CSV or Excel for every record type, with a template that lists the company's own customers, items and accounts as drop-downs; rows are checked before anything is queued.
Record history	An activity timeline on every record: who changed, e-mailed or attached what, and when it reached QuickBooks.

Daily summary	A morning e-mail at 7:00 (Sri Lanka time): yesterday's sales and payments, money owed, bills due this week, items to reorder and approvals waiting. Each user can turn it off.
Comfort	Light and dark theme, works on phones and can be installed as an app, keyboard shortcuts ( shows them all), notifications for approvals and failures.

Changes and approvals

Every create, edit, delete or void becomes an entry in the change queue with one of these states:

STATE	MEANING
pending	Waiting for someone with approve rights for that module.
queued	Approved (or made by a user with direct-write rights); goes to QuickBooks on the next sync, within about a minute.
done	QuickBooks accepted it; the QuickBooks reference number is saved.
failed	QuickBooks refused it, with QuickBooks' reason. It can be fixed and retried, or discarded.
rejected	An approver declined it, with a reason.

The approval inbox shows each change field by field ("Now in QuickBooks" against "Will become"), including line items, and allows approving several changes at once. Edits use QuickBooks' edit sequence, so a record changed in QuickBooks in the meantime is refused instead of overwritten.

Users, roles and access

Permissions are given per module (sales, purchasing, banking, accounting, inventory, lists) and per action: view . create . edit . delete . approve . and direct (write without approval).

Extra permissions: `reports.view` , `sync.run` , `users.manage` , `audit.view` . Roles are edited in the portal; these five come built in:

ROLE	CAN DO
admin	Everything, in every company, including users, roles, companies and the audit log.
accountant	Everything in all six modules including approve and direct write, reports and sync; not users or audit.
sales	View, create and edit sales; view lists; reports. Changes need approval.
purchasing	View, create and edit purchasing; view inventory and lists. Changes need approval.
viewer	Read only, all modules.

People sign in with their **e-mail address or username** and password, then a 6-digit code from **Google Authenticator** (or Microsoft Authenticator / Authy) when two-step verification is on. Each user gets 8 single-use recovery codes. An admin can reset a user's two-step verification when a phone is lost.

Multiple companies

Each Synex Group company is registered once in the portal with its own Web Connector user, password and .qwc file. Its data is kept separate, users are given access per company (admins see all), and people switch company from the top bar.

COMPANY	COMPANY FILE	STATUS
Synex International	SYNEX INTERNATIONAL (PVT) LTD -2025-26.QBW	active, syncing every minute
Synex	abc.qbw (first test setup)	disabled

QuickBooks opens one company file at a time.

Two companies on the same QuickBooks computer cannot sync in the same minute; the one whose file is not open reports "a different company file is open". Keep only the active company in

Web Connector, or use a separate QuickBooks computer per company.

Security

Application

- HTTPS only, with HSTS. Passwords stored as bcrypt hashes; authenticator secrets encrypted.
- **Two-step verification is required** for admins and anyone who can approve or write directly. Until it is set up, they can only reach the Account page.
- Wrong passwords lock the account for 15 minutes after 10 tries, and an address after 30. Wrong authenticator codes lock for 5 minutes after 5 tries.
- Passwords: at least 10 characters with letters and numbers, not a common password, not containing the username.
- Changing a password, a reset by an admin, deactivating a user or "Sign out on all other devices" ends all other sessions at once. Sessions otherwise last 8 hours.
- Content Security Policy, frame blocking, no-sniff and referrer headers; fonts are served by the portal itself, so pages load nothing from other sites. API documentation is switched off.
- Attachments limited to documents and pictures (PDF, images, Office, CSV, ZIP), 20 MB each.
- Audit log of sign-ins (with address), failed sign-ins, every change, approval, export, e-mail and settings change.

Server

- Firewall on; the database listens only inside the server.
 - fail2ban bans an address for 1 hour after 6 wrong SSH sign-ins.
 - Operating-system security updates install automatically; the web server hides its version.
 - Secrets live in a server file readable only by the portal's own system user.
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Hosting and operations

ITEM	DETAIL
Server	Contabo VPS, Ubuntu, shared with other Synex projects.
Address	<code>https://qb.synexint.com</code> (Let's Encrypt certificate, renews automatically). Web Connector uses <code>https://qb.synexint.com/qbwc</code> .
Install folder	<code>/opt/synex-qb-portal</code> , running as its own system user.
Service	<code>synex-qb-portal</code> (systemd). Starts on boot after the database and restarts itself 5 seconds after any crash. <code>nginx</code> , <code>PostgreSQL</code> , <code>fail2ban</code> and <code>cron</code> also start on boot.
Database	<code>PostgreSQL</code> , database <code>synex_qb</code> .
Backups	Nightly at 01:15 a compressed copy kept on the server for 14 days; nightly at 02:00 a copy uploaded to Google Drive with the other Synex databases (kept 7 days on the server).
Source code	Private GitHub repository <code>sathsara-ranasinghe/synex-price-system</code> , branch <code>main</code> .
Settings	Server file <code>backend/.env</code> : database address, JWT secret, first-company Web Connector settings, SMTP e-mail settings, two-step rule (<code>REQUIRE_2FA</code>), daily-summary hour (<code>DIGEST_HOUR</code>).

After a server restart

Nothing has to be started by hand. Allow 2-3 minutes, then open the portal and check that "Synced ... ago" in the top bar moves. Web Connector reconnects by itself; changes made in the meantime wait in the queue and are not lost.

Updating the portal

New versions are tested, committed to GitHub, copied to the server, built there and activated with a service restart, normally with under 10 seconds of downtime.

QuickBooks computer setup

- 1. QuickBooks Desktop 2024 with the company file open, in multi-user or single-user mode.
- 2. QuickBooks Web Connector 2.2 or later.
- 3. In the portal: Companies → company → download the .qwc file.
- 4. In Web Connector: Add an application → choose the .qwc → allow access in QuickBooks ("Yes, always; allow access even if QuickBooks is not running") → enter the company's Web Connector password → tick Auto-Run.
- 5. Keep the computer on and signed in; if it sleeps or Web Connector closes, the portal shows the last synced data and queues changes.

A step-by-step guide with screenshots is in the repository at `docs/QUICKBOOKS_DESKTOP_SETUP.md` .

Technology

PART	BUILT WITH
Web app	Angular 18 with Angular Material 3, installable on phones (PWA).
API and sync	Python FastAPI, SQLAlchemy 2; Web Connector SOAP service and qbXML 16.0.
Database	PostgreSQL 16 with trigram indexes for fast name search.
Documents	ReportLab (PDF), openpyxl (Excel), SMTP for e-mail.
Sign-in	JWT sessions, bcrypt, TOTP (pyotp) with QR setup.
Web server	nginx with Let's Encrypt.

QuickBooks record types are described once in a single registry (fields, field order required by Intuit, which can be created or edited). Lists, forms, import templates, Excel exports and qbXML requests are all generated from that registry, so supporting a new QuickBooks field means changing one place.

Testing and quality

- **434 automated backend tests:** every record type's read, create and edit request against Intuit's rules, the Web Connector conversation, approvals, imports, multi-company isolation, permissions, sign-in and two-step rules, security headers and the newer features.
 - **End-to-end browser test:** starts a throw-away copy of the portal, loads sample QuickBooks data through a simulated Web Connector, then opens every page, all 50 lists, every New form and import page, and fails on any script error or server error.
 - Both suites run before every release.
 - Tested against live data: Synex International (28,893 records on first sync); each list was compared with the database count.
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Limits

- **Not real time:** changes reach QuickBooks in about one minute, and only while the QuickBooks computer and Web Connector are running.
 - **One file at a time** per QuickBooks computer (see Multiple companies).
 - 13 record types cannot be edited through Web Connector; QuickBooks custom fields are read only.
 - Payroll, bank reconciliation, sales tax filing, budgets and closing the books stay in QuickBooks Desktop.
 - Large-data growth: aging and the daily summary load all open invoices and bills; fine at today's size, worth optimising past roughly 100,000 open documents.
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